

Taylor & Maclaurine Series:

Taylor's Theorem

Expand the following functions:

1. $y = \sin x / \cos x$ in the powers of $(x - \frac{\pi}{2})$
2. $y = \text{Log}_e(x)$ in the powers of $(x - 2)$
3. $y = e^{ax}$ in the powers of $(x - 1)$

Maclaurine's Series

Expand the followings:

1. $y = \sin x / \cos x$ in the powers of x
2. $y = \text{Log}_e(x + 1)$ in the powers of x
3. $y = e^x$ in the powers of x .